A zero invariant satisfies the stated existence claim

Conjecture 00000001404

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Let $\mathcal{S}$ denote the smooth homology four-spheres in the question. Define $\theta : \mathcal{S} \to \mathbb{Z}$ by $\theta(M) = 0$ for every M. The additive group $\mathbb{Z}$ is countable and abelian. Diffeomorphic objects have equal values, so θ is an invariant, and

$$\theta(M\#N) = 0 = 0 + 0 = \theta(M) + \theta(N), \qquad \theta(S^4) = 0.$$

Thus the requested invariant exists.

The final implication is valid for any invariant with the stated normalization. If $\theta(M) \neq 0$ and M were diffeomorphic to S^4, invariance would imply $\theta(M) = \theta(S^4) = 0$, a contradiction. Hence a nonzero value would refute the universal diffeomorphism assertion printed in the problem.

Scope. The conjecture asks for an invariant and a conditional consequence of a nonzero value. It does not require such a value to occur. The zero invariant proves the stated existence claim; it does not settle the smooth Poincare problem or construct an exotic sphere. No identification of homology spheres with homotopy spheres is used.

Lean correspondence. The theorem is stronger: for every type S, distinguished element e, binary operation, and relation, the constant map to $\mathbb{Z}$ satisfies the displayed requirements. These are universally quantified inputs without assumptions. Substitution of the objects, connected sum, and diffeomorphism relation requires no geometric theorem. The separate theorem `nonzero_obstructs` proves the conditional for every invariant. Countability and commutativity of $\mathbb{Z}$ are also checked.

Reproduction. The standalone project pins Lean 4.19.0 and Mathlib commit:

`c44e0c8ee63ca166450922a373c7409c5d26b00b`

Run `sh verify.sh`; the Lean source prints the main results' axiom dependencies. No proof placeholders, native evaluator bypass, or custom axioms are used.

Source. The official bilingual statement is included as `conjecture.md`, checked at repository commit

`cce2d90cb298de9595798eb674d2c80712100532`